EXERCISE -2 SQL · AGGREGATES AND OPERATORS

1) SELECT DISTINCT department
from Students;

Department
finance

Department
finance
IT
HR

2) SELECT department, AVG(age) AS avg_age
from Students
GROUP BY department;

Department	AVG - AGE
IT	21
HR	22
finance	23

3) SELECT department, count(*) AS Student_count
from Students
GROUP BY department
HAVING count (*) > 1;

Department	Student_count
IT	2
HR	2

4) SELECT *
from Students
WHERE age BETWEEN 21 AND 22

Student ID	name	age	department
2	Bob	22	HR
3	charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

5) SELECT *
from Students
WHERE (department='IT' OR department='HR')
AND age > 21;

Student ID	name	age	department
2	BOB	22	HR
5	Eve	22	HR

6) SELECT department, Sum(credits) As total_credits
from Courses
GROUP By department
HAVING Sum(credits) > 5;

Department	Total_Credits
IT	11

7) SELECT *
from Courses
WHERE not Credits = 4;

Course id	Course name	department	Credits
101	SQL Basics	IT	3
104	Excel	finance	2
105	Statics	HR	3

8) SELECT course id, coursename, Credits,
from Courses
ORDER By Credits DESC
Limit by 3;

Course id	Course name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

SECTION -3 ENROLLMENTS TABLE

9) SELECT max_grade AS max-grade,
min_grade AS min_grade,
Av_grad AS avg-grade
from enrollments;

max grade	min grade	Avg_grade
90	78	84.6

10) SELECT course_id
Count (enrollments_id) As enrollment_count
from enrollments
GROUP By course_id;

Course id	enrollment count
101	1
102	1
103	1
104	1
105	1

11) SELECT depart,
Sum (total salary) As total_salary
Sum(total_bonus) AS total-bonus
from Salaries
GROUP By depart;

department	total salary	Total Bonus
IT	122 000	10,500
HR	117 000	7500
finance	70000	6000

15) SELECT
from projects
WHERE projects = budget BETWEEN 50 000
AND 120 000 AND (department) NOT Marketing.

Project_ID	Project_name	department	Budget
1	AI APP	IT	120000
2	Payroll system	finance	80 000
5	HR Portal	HR	50 000